

18 Up and Down Sampling

Lets think about continuous to digital converters some more.

If we have an $X(j\omega)$ in continuous time (i.e. for example a box $u(\omega - \omega_0)u(-\omega - \omega_0)$), then we can take it to discrete time Fourier representation by rescaling it: divide the height by T , stretch the frequency axis so that ω_0 maps to $\omega_0 T$, and tile it to be periodic every 2π .

This applies for any signal. To go from discrete time to continuous, we just do the inverse - keep only the representation from $-\pi$ to π squish the frequency axis to map $\omega_0 T$ to ω_0 , divide the amplitude by T .

Side note:

If you have high frequency data, that's good for bandwidth and time resolution in the frequency domain. However, it takes a lot of resources to store all that data, so we really only want to sample as fast as useful.

Down Sampling

Decreasing your sample rate by an integer factor N .

We have a discrete time signal, multiply by a discrete time pulse train, pass it through a "decimation block" of a factor N , and take out our new x_b signal.

The pulse train $p[n]$ is a delta every N samples, and zero everywhere else. This keeps only time indices that are integer multiples of N .

The decimation block discards the zeros in between. We take every n^{th} sample, and make them adjacent to each other in the signal.

What is happening in the frequency domain when we do this?

Time		Frequency	
$p[n]$	$\sum_k \delta[n - kN]$	$P(e^{j\hat{\omega}})$	$\sum_{k=0}^{N-1} \frac{2\pi}{N} \delta(\omega - k \frac{2\pi}{N})$
$x_p[n]$	$x[n]p[n]$	$X_p(e^{j\hat{\omega}})$	$\frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - \frac{2\pi}{N}k)})$ and $\sum_{k=-\infty}^{\infty} x_p[kN] e^{-jk\omega N}$
$x_b[n]$	$x_p[Nn]$ and $x[Nn]$	$X_b(e^{j\hat{\omega}})$	$X_p(e^{j\frac{\hat{\omega}}{N}})$

The key take away: When down sampling in software, we have to filter and then decimate. Otherwise, we'll have aliasing.

The derivation of this table is found below.

Lets do the time domain analytical expression first:

$$p[n] = \sum_k \delta[n - kN]$$

$$P[n]$$

The discrete Fourier series

$$\begin{aligned}
 p[n] &= \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}kn} \\
 a_k &= \frac{1}{N} \sum_{n=0}^{N-1} p[n] e^{-j\frac{2\pi}{N}kn} \\
 &= \frac{1}{N} \forall k
 \end{aligned}$$

The discrete time Fourier transform

$$\begin{aligned}
e^{j\omega_0 n} &\leftrightarrow 2\pi\delta(\omega - \omega_0) \\
e^{j\frac{2\pi}{N}kn} &\leftrightarrow 2\pi\delta\left(\omega - \frac{2\pi}{N}k\right) \\
\sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}kn} &\leftrightarrow 2\pi \sum_{k=0}^{N-1} a_k \delta\left(\omega - \frac{2\pi}{N}k\right)
\end{aligned}$$

so

$$p[n] = \sum_{k=0}^{N-1} \frac{1}{N} e^{j\frac{2\pi}{N}kn} \leftrightarrow \sum_{k=0}^{N-1} \frac{2\pi}{N} \delta\left(\omega - k\frac{2\pi}{N}\right)$$

So the frequency domain of our pulse chain is deltas of height $\frac{2\pi}{N}$ spaced every $\frac{2\pi}{N}$.

x_p

There are two approaches we could do.

Approach 1, convolution

$$x_p = x[n]p[n] \leftrightarrow X_p(e^{j\hat{\omega}})$$

$$\begin{aligned}
X_p(e^{j\hat{\omega}}) &= \frac{1}{2\pi} P(e^{j\hat{\omega}}) * X(e^{j\hat{\omega}}) \\
&= \frac{1}{2\pi} \int_{2\pi} P(e^{j\theta}) X(e^{j(\omega-\theta)}) d\theta \\
&= \frac{1}{2\pi} \int_{2\pi} \frac{2\pi}{N} \sum_{k=0}^{N-1} \delta\left(\theta - \frac{2\pi}{N}k\right) X(e^{j(\omega-\theta)}) d\theta \\
&= \frac{1}{N} \sum_{k=0}^{N-1} \int_0^{2\pi} \delta\left(\theta - \frac{2\pi}{N}k\right) X(e^{j(\omega-\theta)}) d\theta \\
&\text{this is the sifting property} \\
&= \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - \frac{2\pi}{N}k)})
\end{aligned}$$

We are scaling by $\frac{1}{N}$ and have a bunch of copies of our $X(e^{j\hat{\omega}})$ centered at integer multiples of $\frac{2\pi}{N}$.

The $x_p(e^{j\hat{\omega}})$ is periodic every $\frac{2\pi}{N}$ with a height $\frac{1}{N}$ of our $X(e^{j\hat{\omega}})$

If we have aliasing, then the boundaries between these would overlap - so the width of the new version would be enough to touch from either side. Aliasing can happen not only when we go from continuous to discrete time, but any time that we change our sampling rate.

Approach 2: Definition of DTFT

$$\begin{aligned}
x_p[n] &= \sum_k x[kN] \delta[n - kN] \leftrightarrow X_p(e^{j\hat{\omega}}) \\
\delta(n - kN) &\leftrightarrow \sum_{n=-\infty}^{\infty} \delta[n - kN] e^{-j\omega n} \\
&= e^{-j\omega kN}
\end{aligned}$$

This is a linear operator, so we have

$$\begin{aligned}
x_p[kN] &\leftrightarrow X_p(e^{j\hat{\omega}}) \\
\sum_{k=-\infty}^{\infty} x[kN] \delta(n - kN) &\leftrightarrow \sum_{k=-\infty}^{\infty} x_p[kN] e^{-jk\omega N}
\end{aligned}$$

$x_b[n]$

$$\begin{aligned} X_b(e^{j\omega}) &= \sum_k x_b[k]e^{-j\omega k} \\ &= x_p[kN]e^{-j\omega k} \\ &= \sum_{n=\text{integer multiple of } N} x_p[n]e^{-j\omega \frac{n}{N}} \\ &\text{because } x_p[n] = 0 \text{ when } n \neq kN, k \in \mathbb{Z} \\ &= \sum_{n=-\infty}^{\infty} x_p[n]e^{-j\omega \frac{n}{N}} \\ &= X_p(e^{j\frac{\omega}{N}}) \end{aligned}$$

So we're only stretching the frequency axis by a factor of N . When in $X_p(e^{j\hat{\omega}})$ having a width ω_0 , the width becomes $N\omega_0$. Instead of being centered at $\frac{2\pi}{N}$, now they are centered at 2π . The height is the same as before, $\frac{1}{N}$.

Our results are analogous to continuous time. Aliasing occurs if $\omega_0 > \frac{\pi}{N}$. If we will have aliasing, we'll have to apply an anti-aliasing filter before down-sampling.

Up-sampling

Increasing the sampling rate by integer factor N .

We do this by padding zeros and interpolating.

$$x_b[n] \rightarrow \text{zero padding} \rightarrow x_p[n] \rightarrow \text{interpolation envelope} \rightarrow x[n]$$

We're doing the same operation as before, but in the reverse.

